Building Code Requirements

- The deck shall be positively anchored to the primary structure and designed for both vertical and lateral loads. Such attachment shall not be accomplished by the use of toenails or nails subject to withdrawal.

  IRC 2018/2021/2024 Section R507.8, R507.9

- Where the lateral-load connection is permitted to be installed per Figure R507.9.2(1), holdown tension devices shall be installed in not less than two locations per deck, within 24" of each end of the deck, and have an allowable stress design capacity not less than 1,500 lb.

  IRC 2018/2021/2024 Section R507.9.2

- Where the lateral-load connections are provided in accordance with Figure R507.9.2(2), the holdown devices shall be started within 24 inches of each end of the deck and have an allowable stress design capacity of not less than 750 lb. **Note:** a minimum of four holdowns are required evenly distributed along the deck in this configuration.

  IRC 2018/2021/2024 Section R507.9.2

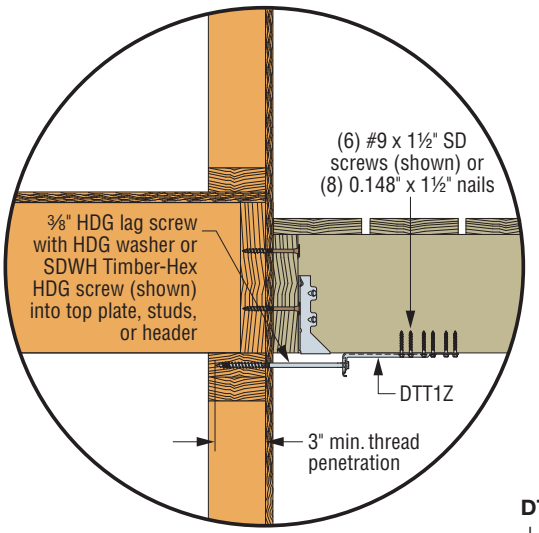
Lateral Support at Band Joist

For decks that are partially supported by an adjacent structure (such as a house), the connection between the deck and that structure is vital. A bolted or screwed ledger-to-rim board connection is suitable to support gravity loads; however, in some cases the building codes require a connection that is able to resist lateral loads. In these situations, tension ties are typically called out to tie the joists of the deck directly to the structure.

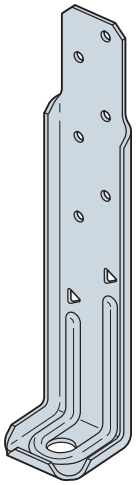
DTT™ deck tension ties are a safe, cost-effective solution designed to meet or exceed lateral-load code requirements for deck construction.

DTT1Z® satisfies the IRC provision for a 750 lb. lateral-load connection to the house at four locations per deck. This code detail permits the lateral connection from the deck joists to be made to top plates, studs or headers within the supporting structure.

DTT2 can be used to satisfy the IRC provision for a 1,500 lb. lateral-load connection at two locations per deck.

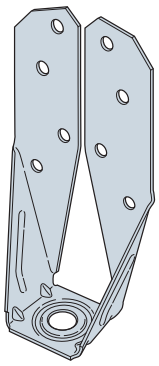


IRC 2018/2021/2024 Figure R507.9.2(2)
Typical DTT1Z Deck-to-House Lateral-Load Connection

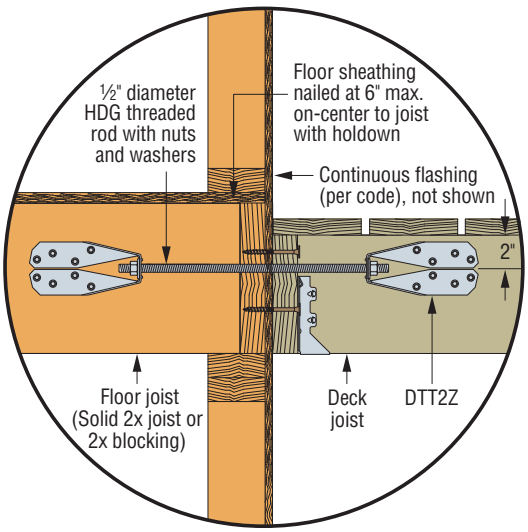


DTT1Z Deck Tension Tie
US Patent 10,865,588B2

DTT1Z kit includes Strong-Drive SDWH Timber-Hex HDG and SD Connector screws.



DTT2 Deck Tension Tie
(available in ZMAX® and stainless steel)
US Patent 8,555,580B2



IRC 2018/2021/2024 Figure R507.9.2(1)
Typical DTT2 Deck-to-House Lateral-Load Connection

For more information on lateral-load connections, refer to Simpson Strong-Tie technical bulletin T-C-DECKLAT at **strongtie.com**.